

CANONICAL LESSON

Build the Evidence Contract

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C02
Notebook	V7_0_N02
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Specify what every record means and which transformations, linkages, access conditions, and retention rules are permitted.

Learning outcomes

- Define entities, identifiers, measures, classifications, reference periods, spatial grain, provenance, lawful access, linkage rules, and retention before analysis.

Why this matters

Successful joins can create false evidence when concepts, periods, populations, or lawful purposes are incompatible.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

1. Evidence is a contract

- A table is not self-explanatory.
- The same column name can represent different concepts across agencies.
- A usable source contract states what each record means, who created it, when it applies, and which transformations are permitted.

2. Validate identifiers and grain

- A valid key must be unique at the declared entity-time-space grain.
- Failure means aggregation, duplication, or linkage may produce false counts.

3. Classification and unit controls

- Values become comparable only after concepts, units, code lists, and versions are aligned.
- Never silently map an unknown code to “other”.

4. Provenance ledger

- Every transformation should have an input, rule, output, actor, time, and version.
- A provenance hash detects later change; it does not prove the data are true.

5. Linkage is a governed operation

- Joining datasets can create new sensitivity and new error.
- Specify join cardinality, permitted fields, match review, unresolved cases, and whether the linked output may persist.

6. Data minimization and retention

- Collect the least detail needed for the decision.
- Retention requires a purpose, duration, custodian, deletion/archive trigger, and audit record.

Worked cross-sector example

A rainfall station contract binds the identifier, coordinates version, unit, observation time, calibration status, provenance, and quality owner.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Draft a source contract for rainfall stations.
- 2.
- State the key and grain for monthly facility reporting.
- 3.
- Explain why a successful join may still be invalid.
- 4.
- Define a retention schedule for a linked analytical table.

Answer key

- 1.

- Include station entity, identifier, coordinates/version, variable/unit, observation time, authority, calibration, update cycle, access, retention, and quality owner.
- 2.
- A candidate key is facility identifier + reporting month, subject to the declared service/disaggregation grain.
- 3.
- Keys can match while concepts, periods, populations, authority, or quality are incompatible.
- 4.
- State purpose, lawful basis, approved users, active duration, review date, deletion/archive rule, and evidence of disposal.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.